

Amanah AI

AI-Powered Verification and Correction of Quranic and Prophetic Quotations

Project documentation prepared for the evaluation committee, organised according to the published selection criteria.

Track	IslamicEval 2025 Subtask 1: quotation detection (1A), verification (1B), correction (1C)
Live demo	https://ghada-99-ragab-amanah-ai.static hf.space
Source code	https://github.com/Ghada-Ragb/Amanah_AI
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1. Executive summary

Language models quote the Quran and the Hadith fluently, and sometimes wrongly: a word is replaced, two verses are spliced, a hadith is invented or attributed to the wrong speaker. A reader cannot tell the difference, and in religious content a confident mistake is worse than no answer.

Amanah AI is a web tool that finds every Quranic verse and Hadith quotation in a text, compares each one word by word with the reference corpora, corrects Quranic wording verbatim from the source text, and refers anything uncertain to a human reviewer. The system never invents a verse, hadith, reference or correction: when the evidence is not sufficient, it says so.

Capability	What the user gets
Detection	Every verse and hadith in the text is located, with or without quotation marks
Verification	Each quotation is matched word by word with the Quran and the six canonical Hadith books
Correction	Altered Quranic wording is replaced with the exact source text, with one-click copy
Safe abstention	Uncertain cases are referred to human review instead of being guessed
Private by design	Verification runs inside the user's browser; the text is not sent to a server
Two ways to use it	Ask & Verify (question, answer, check) and Verify directly (paste any text)

2. Clarity of the problem and relevance to the track and audience

2.1 The problem

Quranic text must be reproduced exactly, and a Hadith belongs to a specific narrator and book. Four error patterns recur in model answers, and each one can be reproduced in the live demo:

- **Substituted word.** A single word is changed, for example «ثم استقم» instead of «فاستقم».
- **Spliced or reordered verses.** Parts of two ayahs are merged, or the word order is altered.
- **Invented Hadith.** A fluent text with no entry in the canonical books.
- **Wrong attribution.** An ayah presented as a saying of the Prophet, or a hadith presented as divine speech.

2.2 Audience

Preachers and writers who draft with AI assistance, students and teachers, publishers and moderators of Islamic platforms, and developers of Arabic assistants who need a safety layer. Each of them needs a verdict they can trust, with the evidence and a plain reason.

2.3 Fit with the track

The tool implements the three IslamicEval 2025 Subtask 1 tasks end to end on the official data: detection of quotations (1A), verification of their correctness (1B) and correction of wrong ones from the official sources (1C). Three clear outcomes are shown: verified, mismatch (with the exact source text for the Quran) and needs human review.

3. Suitability of the AI approach and the added value

Each component is given the job it can do reliably:

Component	Technique	Why it fits
Detection (1A)	Quotation-mark and bracket rules, a corpus scanner for unannounced quotations, and an optional CAMELBERTE-MSA token classifier	Corpus-backed spans take priority; the learned model can add evidence, never override it
Retrieval	BM25 over the pre-tokenised corpora, re-ranked by character n-gram similarity	Fast, explainable, and light enough to run in the browser
Alignment	Word alignment on a phonetic skeleton (sliding window, LCS, dynamic gap)	Pinpoints the exact altered word
Decision (1B)	Explicit rules: verified, corrected, unsupported, human review	Reproducible and testable
Question answering	A language model through a secure proxy (Ask & Verify page)	Produces the answer to be checked; it never judges correctness

This is retrieval-augmented verification used where it matters: the retrieved record is not used to generate text, it is used to judge and to copy from. In a domain where generation is the source of the problem, the source text is the authority.

Added value: in seconds, inside the browser, the user receives a colour-coded report with the cited evidence, a plain-language reason and the corrected Quranic text ready to copy.

4. Reliability and scientific-integrity plan

4.1 Anti-hallucination rules

Rule	How it is enforced
Corrections come only from the source	A correction is the verbatim corpus text. A grounding guard removes any correction or suggestion that is not a normalised substring of the cited record.
Evidence must exist	A cited source must exist in the corpus. An unsupported quotation carries no invented evidence.
Hadith are never auto-corrected	A Hadith is verified only on an exact word match. Any added, missing or changed word goes to human review, because a word can change a ruling.
Very short quotes	Quotations of fewer than three words go to human review unless they are exactly a whole ayah.
Final verdict	A quotation is reported as correct only when it is verified against the source.
Responsible scope	The tool checks text against sources. It is an aid for textual checking and does not replace specialist review.

4.2 Human review as a designed outcome

Referring a case to a person is a first-class result of the system, not an error. This is what allows the tool to be conservative where religious accuracy is at stake.

4.3 Testing and evaluation

- A comprehensive automated test suite covers normalisation, detection, alignment, decisions, grounding, Hadith strictness, every saved example, proxy prompt parity, notebook consistency and a secret scan.
- Evaluation on the official IslamicEval 2025 development data, reported against simple baselines for verification and correction.
- A synthetic stress test of fabricated, spliced, substituted, dropped and reordered quotations confirmed that fabricated or spliced text is not accepted as verified.
- A reproducible Colab notebook rebuilds the evaluation, the examples and the stress test.

4.4 Data provenance and licensing

Resource	Licence	Handling
Quran text (Tanzil)	CC BY 3.0	Used verbatim and unmodified; credit and link shown in the interface
IslamicEval 2025 repository and data	Apache-2.0	Cited; the development subset is used for evaluation

Resource	Licence	Handling
Hadith compilation (OmarShafie/hadith)	Code: GPL-3.0; texts not separately licensed upstream	Disclosed in NOTICE.md and the data-sources document
Project source code	MIT	Copyright Amanah AI contributors

Corpus files were verified to be identical to the upstream files. No secret, key or real user data is stored in the repository or the page; the model proxy limits input size, origins and request rate.

5. Feasibility within the challenge period

The solution is complete and running.

- A static site on Hugging Face Spaces: Python runs in the browser through Pyodide in a Web Worker, so verification needs no server.
- A Cloudflare Worker proxy (Workers AI free tier) serves the question step of the Ask & Verify page.
- Fast start: the engine and the Quran index load first, and the Hadith index loads in the background.
- Reproducible: build, test and evaluation commands are documented, and a Colab notebook runs the full pipeline.
- Sustainable: current running cost is zero; the plan covers hosting and model usage through free tiers, partnerships and team resources.

6. Originality and added value

- **Corrections are copied, not generated.** The corrected verse is the exact corpus text, and the grounding guard makes a fabricated correction structurally impossible.
- **Abstention by design.** The system says «a person must decide» when evidence is not enough.
- **Strict Hadith policy.** A changed word is sent for review instead of being silently repaired.
- **Unannounced-quotation detection.** Quotations without quotation marks are found by scanning the corpus.
- **Runs in the user's browser.** No verification text leaves the device, and there is no verification server to maintain or secure.
- **Transparent and reproducible.** Sources, licences, tests and the evaluation pipeline are all documented and open.

7. Ability to execute and to cover the required tasks

Requirement	Delivered
Live solution link	https://ghada-99-ragab-amanah-ai.static.hf.space
Public repository with the complete code	https://github.com/Ghada-Ragb/Amanah_AI
Runtime files without secrets	Included; a secret scan is part of the test suite
Project documentation	This document, plus README, DOCUMENTATION, DEPLOYMENT, REFERENCES and DATA_SOURCES_AND_LICENSES in English and Arabic, and NOTICE.md
Presentation	A deck prepared on the official template
Explanatory video	Submitted with the entry
Tasks 1A, 1B and 1C	All three implemented and evaluated on the official development data

The team divided the work between AI engineering (detection, retrieval, alignment, evaluation) and user-interface engineering (Arabic interface, static build and deployment).

8. Using the system

Page	Steps
Ask & Verify	Type a question and press «اسأل ثم تحقق». The model answers, then every verse and hadith in the answer is checked. Press «جرب مثلاً» for a saved scenario.
Verify directly	Paste any text and press «تحقق من النص». Press «جرب مثلاً» to load one of the saved scenarios covering every combination of correct and incorrect verses and hadiths.
Reading the result	Green: verified. Red: mismatch, with the exact source text for the Quran and a copy button. Amber: needs human review.

9. Repository map and how to run

Path	Purpose
verifier.py, alignment.py, retrieval.py, scanner.py, detector.py	Verification engine
ui.py, app.py, build_static_space.py	Arabic interface, local Gradio app, static site builder
worker/, server/	Cloudflare Worker proxy and shared answer sanitiser

Path	Purpose
data/, index/	Corpora and prebuilt retrieval indexes
tests/, evaluate.py, research/	Tests, evaluation and stress test
colab_verification_pipeline/	Colab notebook and run guide
docs/	Documentation, deployment, references and data licences

Run locally: `python -m unittest discover -s tests` | `python evaluate.py` | `python build_static_space.py`

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